



Instructor: Date:

Learner: Regional Branch:

Program: Center:

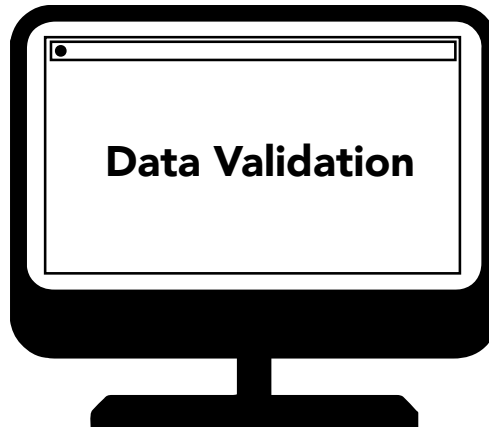
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

The Importance of Data Validation



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain the concept of data validation.
- I can talk about additional features related to data validation.
- I can give examples of data validation.

Ask your partner:

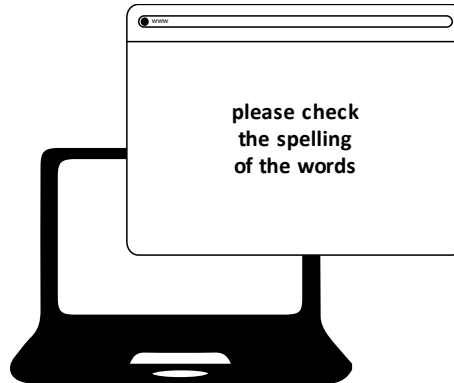
- Do you think it is important to set limitations to what a program can do? Why?
- Why do you think it is important to validate data in an entry box?





Let's Explore!

Let's validate this Data!



Instructions: Read the text below.

Peter: Today, we're going to start this session by discussing what data validation is for. So, any volunteers?

Mark: Well, If not mistaken, we can use data validation processes to ensure that the kind and length of characters that people input in a text box correspond to the ones we expect.

Peter: Excellent. For example, right now, I'm trying to type numbers, letters, and symbols into this text entry box that only accepts numbers. The program by means of the data type validation process is going to verify those numbers and symbols to check if they are valid or allowed; notice the program is not letting me do so, because of the restriction. This process, as you can see, involves two different steps: the validation check that verifies if the data is valid or not and the post check action, where the post validation process sends feedback to correct possible errors.

Mark: Excuse me, but does that mean that we can use data validation processes to restrict the input of unwanted data?

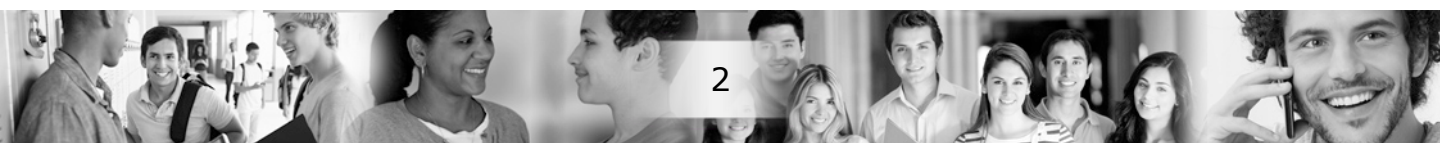
Peter: Of course, if you set the proper restrictions, the data validation process gives you accuracy and consistency of data input to apps and systems.

Mark: Ok. That's interesting. Would you mind giving me some more examples?

Peter: In computer science, data validation may warrant a program works correctly in accordance with the data you provide. But, you must be sure about the restrictions you set when you are coding the program; you must also know that these restrictions are the same validation rules. If you know how to work with them, such rules are going to provide you with security in all the data that you input to the system.

Mark: Thanks for the explanation.

Peter: Next session we are going to learn how to work with data validation processes in programs, such as Excel.



**Let's Practice!**

Do you know what data validation is?

Instructions: Use the words from the word bank to complete each sentence.

Word Bank

verifying / going to confirm / confirmation / carrying / verifies

1. By means of data validation processes you are _____ that the character you input in the text box corresponds to the character you expect.
2. For example, In a cellphone, the data type validation process _____ the numbers or symbols you enter.
3. As you see the software is _____ out two different steps to validate the data, the validation check and the post check action.

**Let's Practice!**

You can proofread sentences in software documents

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: can limit, must be, should verify.

1. When you use data validation, it is important to be sure that the information you're entering _____ totally acceptable.
2. The programmer _____ the amount of characters you want people to enter in the text box.
3. You _____ what kind of restrictions you are setting on that text field.



Extension Activity

Now is your turn

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find out information about the concept of data validation.
- Write down all the features related to data validation you find relevant to work with a database.
- Share your findings to your classmates.

Self-Assessment

Yes

No

- I do further research on the concept of data validation.
- I take notes of additional features related to data validation and databases.
- I share my findings with my classmates.

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Target Words

- accuracy** (adjective): something appropriate
e.g. If you set the proper restrictions, the data validation process gives you accuracy and consistency of data input to apps and systems.
- validation rules** (noun): a set of guidelines to validate something
e.g. You must also know that these restrictions are the same validation rules. If you know how to work with them, such rules are going to provide you with security in all the data that you input to the system.
- warrant** (verb): to ensure; to guarantee
e.g. In computer science, data validation may warrant a program works correctly in accordance with the data you provide.

- text entry box** (noun): a rectangular area in any program where you can add text
e.g. Excellent. For example, right now, I'm trying to type numbers, letters and symbols into this text entry box that only accepts numbers.
- character** (noun): a symbol that represents a part of speech such as letters or numbers
e.g. We can use data validation processes to ensure that the kind and length of characters that people input in a text box correspond to the ones we expect.
- restriction** (noun): a limitation
e.g. You must be sure about the restrictions you set when you are coding the program.



Learning Points

Simple Present, Present Continuous, and Be Going to, to Explain What Data Validation Means

Present simple, present continuous and be going to are time tenses we use to express our ideas depending on specific circumstances.

PRESENT SIMPLE

Use **present simple** forms to refer to routines and general ideas that take place at a general time or with some sort of frequency.

To conjugate a verb in present simple you simply add an "s" or "es" to the third person singular, depending on the verb.

1st person	<i>I</i>	<i>program software.</i>
2nd person	<i>You</i>	<i>program software.</i>
3rd person	<i>He, She, It</i>	<i>programs software.</i>
Plural forms	<i>We, You, They</i>	<i>program software.</i>

Examples:

*I **work***

*She **works***

*I **study***

*He **studies***

Use the following structure to make sentences in present simple.

Subject + Verb + complement.

Example:

*She **works** for a software company.*

Use the following structure to make questions in present simple.

Do/Does + subject + verb + complement + ?

Example:

***Do** you **know** any good program to validate data?*

***Does** the IT engineer **know** how to validate data?*

Use the following structure to make negative statements in present simple.

Subject + do/does not + verb + complement.

*I **don't have** any idea about programming!*

*She **doesn't come** to the company on Mondays.*



Learning Points

Note that regardless the person in interrogative and negative forms, the verb is without conjugation. We call that a "bare infinitive or infinitive *without to*."

IMPORTANT:

For any other verb different from the verb to be, we use the auxiliary verb "**do**" to make questions and negative statements.

Compare:

Do you **know** how to validate data with Excel?

Are you happy?

Does he **work** here?

Is she here?

PRESENT CONTINUOUS

We use **present continuous** forms to express ongoing actions, that means actions that are taking place at the moment of speaking or actions that are temporarily happening in the present.

Use this structure to make sentences in present continuous form:

Subject+ (Aux) be + main verb+ ing+ complement

Example:

- He **is calling** his manager right away.
- She **is working** in an important project now.

In order to make questions you use the following structure:

(wh) + be + subject + ing + complement + ?

Example:

What **are** you **doing** with that computer?

Are you **working** on the data validation project?

BE GOING TO

You can use "**be going to**" to talk about plans in the future based on prior decisions or to make predictions based on present evidence.

The structure is very simple:

Subject + be going to + verb + complement.





Learning Points

Example:

We **are going to have** a meeting later.

Diane **is going to be** in charge of the project.

In its negative form we use the following structure:

Subject + be + not + going to + verb + complement.

Example:

Shannon **is not going to finish** that project for tomorrow.

I'm **not going to fix** the program if I don't know what the problem is.

For the interrogative form we use the following structure:

(Wh) + Be + Subject + going to + verb + complement + ?

Examples:

Why is she **going to be** the new leader?

Are you going to implement that code in the system?

Examples

- Does that mean that we can use data validation processes to restrict the input of unwanted data?
- As you can see, the program is not letting me do so, because of the restriction.
- The data type validation process is going to verify those numbers and symbols to check if they are valid or allowed.

Infinitives of Purpose Review

You can use the **infinitive form of the verb** (to) to show the purpose of an action.

You can use the infinitive of purpose to show the reason why someone does something.

The infinitive of purpose shows the reason why the subject of the sentence does what he/she does; therefore, if the subject is different from the one showing the purpose you use **"so that"**

Compare:

Jane is coding the program **to validate** all the data we have in the server.

Jane is coding the program **so that** the IT department **validates** the data we have in the server.

If you notice, in the first example, the reason why jane is coding the program is to validate data, but in the second example, the reason why she is coding the program is for other people to validate the data.





Learning Points

In a more formal context you can use *in **order to*** or ***so as to***

Compare:

*Johnna works hard **to make** his family happy.*

*The IT department works hard **in order to make** the data validation tools be up to date.*

*Both companies work together **so as to safeguard** their information.*

Examples

- *We can use data validation processes to ensure that the kind and length of characters that people input in a text box correspond to the ones we expect.*
- *The program by means of the data type validation process is going to verify those numbers and symbols to check if they are valid or allowed.*
- *Use data validation processes to restrict the input of unwanted data.*

List of programming and computer science terms

In programming, you can find different computer programs and useful computer science terms that are good to know beforehand to understand how programs work.

Debugger:	It is a computer program whose purpose is to review, test, and fix bugs of other programs.
Decompiler:	It is a computer program that translates a low level code of abstraction, such as computer language, to a higher level of abstraction in executable programs.
Defragment:	The process of putting all pieces of a disk back together, as the OS commonly stores information in every empty space it finds.
Delphi:	It is a programming language that aims to build native applications for Linux and Microsoft.
Device driver:	Computer program that allows the OS recognize and interact with the hardware peripherals, such as printers, cameras, or modems.